

Tutorial 9

Answers

Problem 10-1

$$X_{CG} = 8.774 \text{ in}$$

$$I_{zz} = 0.394 \text{ lb-in-sec}^2$$

$$K = 9.354 \text{ in}$$

Problem 10-4

$$a. \begin{bmatrix} -5 & -2 & 12 & -1 \\ 1 & 3 & -2 & 4 \\ -1 & -1 & 1 & 0 \\ 3 & -3 & 7 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -9 \\ 10 \\ -7 \\ -6 \end{bmatrix}$$

$$b. \begin{bmatrix} 3 & -5 & 17 & -5 \\ -2 & 9 & -14 & 6 \\ -1 & -1 & 0 & -2 \\ 4 & -7 & 8 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -5 \\ 22 \\ 13 \\ -9 \end{bmatrix}$$

Problem 10-22

$$I_{eff} = 1.506 I_{diam}$$

Problem 10-24

$$m_{eff} = 4.5 \text{ kg}$$

$$K_{eff} = 888.9 \text{ N/m}$$